

COMP5566 Project Presentation

Quantifying Frontrunning Activities on Ethereum and Their Impacts on Marketplace

Group 1

1. Project Brief & Requirements

Goal: Collect historical Ethereum DEX transactions, detect frontrunning patterns, quantify adversary profits, and measure market impact on Uniswap V2.

Steps

1. Collect historical UniswapV2 logs
2. Detect Frontrunning activities (Displacement, Insertion, Suppression and Arbitrage)
3. Analyze the profits of adversaries

Pipeline

Dataset Crawler → 4 MEV Detectors → Profit Analyzer → Charts + Excel

2. The Data

2.1 How we do this

- Dune is paid, and we're not rich.
- Etherscan API is suitable.
 - Free quota
 - Filter `Txs` `logs` based on contract address
 - Has rate limit by we can use multiply accounts for load balance.

2.2 What we have done

- We fetch five popular pairs:
 - `USDT-USDC` `WETH-DAI` `WETH-USDC` `WETH-USDT` `WETH-WETC`
- Every single from Uniswap's begining till March, 2026 was downloaded
 - accross **14,639,489** blocks
 - retrieve **2,690,493** logs
 - Use **~8 hours** for smooth collect
 - Uploaded to <https://www.kaggle.com/datasets/chickenbilibili/uniswapv2-exchange-history>

3. The Detail of collected Data

- **Who are the real dealer?:** if sender is UniswapV2 router, use `to` address instead
- **Key records:** `block_number` + `timestamp` + `tx_index` + `log_index` + ...
- **Event** We examined `UniswapV2Pair.sol` and collect four event: `Swap` / `Sync` / `Mint` / `Burn`

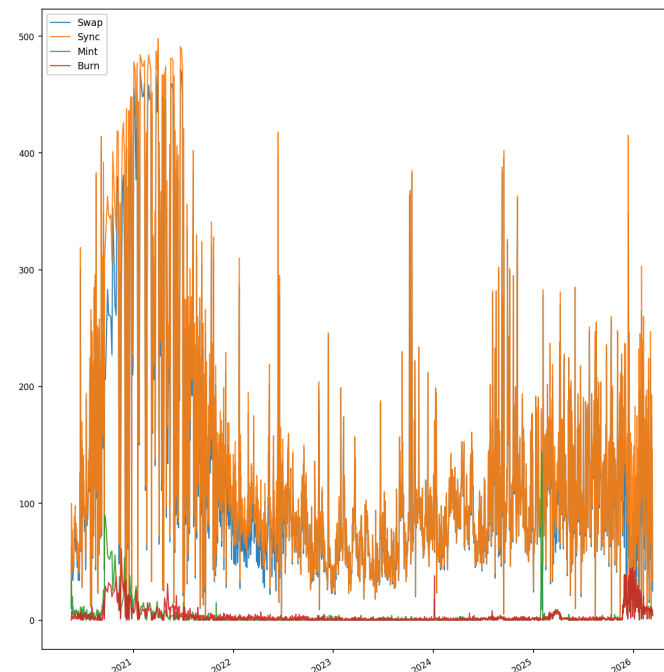
By event type

Logs	records
Sync	1,349,490
Swap	1,315,097
Mint	13,381
Burn	12,505
Summary	2,690,473

By pair

pair	records
WETH_USDC	635,388
WETH_DAI	569,821
WETH_USDT	566,975
USDC_USDT	493,501
WETH_WBTC	424,788
Summary	2,690,473

Daily log density (WETH-WBTC)



4. Detection Methods

Type	Logic	Key Threshold
Sandwich (Insertion)	Buy before victim, sell after — profit from victim's price impact	victim same direction as front-run
Displacement	Pay higher gas to execute the same trade before victim	gas ratio $\geq 1.5x$
Arbitrage / Back-run	Trade opposite direction right after a large swap to capture price gap	trade size $> P90$, within 3 tx
Suppression	Flood block with many high-gas txs to crowd out other users	≥ 3 swaps, gas $\geq 3x$ median

Profit models ($G = \text{gasPrice} \times 150000 / 10^{18} \times P_{ETH}$):

Sandwich: $n_i = \sum \text{out}_i - \sum \text{in}_i$, $\text{Net} = \text{USD}(n_0) + \text{USD}(n_1) - G_f - G_b$

Displacement: $\text{Loss} = \text{AMM}(\text{in}_v, R_{\text{before}}) - \text{out}_v$, $\text{Net} = \text{USD}(\text{Loss}) - G_f$ (AMM = Uniswap V2 $xy = k$ with 0.3% fee)

Arbitrage: $\text{Net} = \text{USD}_{\text{pre}}(\text{out} - \text{in}) - G_b$ (use previous block price as fair reference)

5. Overall Detection Results

Type	Events	Key Metric	Proportion
Sandwich	845 (16.2%)	Net profit: \$194,985	21.2% profitable
Displacement	758 (14.5%)	Est. profit: \$575,132	15.3% profitable
Arbitrage / Back-run	3,593 (68.9%)	Net profit: \$9,895,544	avg \$2,754/event
Suppression	18 (0.3%)	Median gas premium: 20.5x	extreme but rare
Total	5,214	\$10.7M+ aggregate	

- Arbitrage / back-running dominates both in **frequency** and **total profit**
- Sandwich profitability is **highly concentrated** — most attempts lose money on gas
- Suppression is the rarest but shows **extreme** gas competition (median 20.5x)

6. Sandwich Attack Findings

- **845** events, only **179** profitable (**21.2%**) — profitability highly concentrated
- Gross profit: **\$668,920**; Net profit (after gas): **\$194,985**
- Avg net: \$231; Median net: **-\$23** (most attackers lose money on gas)

7. Displacement, Arbitrage and Suppression

Displacement (758 events) — ordering advantage via gas premium

- Median gas ratio: **3.2x** (P95: 125x); Top frontrunner: **82** events

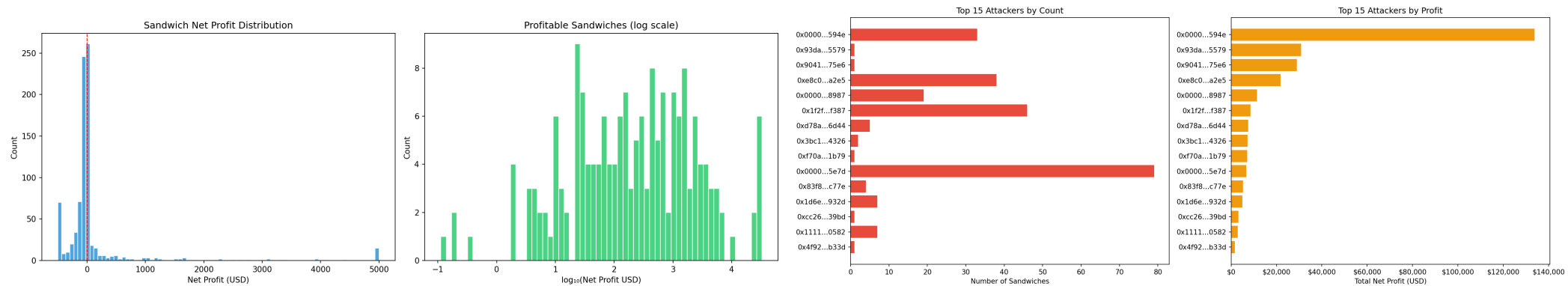
Arbitrage / Back-running (3,593 events) — most frequent pattern

- Net profit: **\$9,895,544** (avg **\$2,754** per event); Top back-runner: **210** events

Suppression (18 events) — rarest but most extreme, median gas premium: **20.5x**

8. Profit Distribution & Top Attackers

- Profit is **extremely concentrated** — a power-law distribution
- Top **10** back-run events account for **85.1%** of total arbitrage profit (\$8.4M / \$9.9M)
- Top **50** events account for **94.3%** — the remaining **3,543** events share only **5.7%**

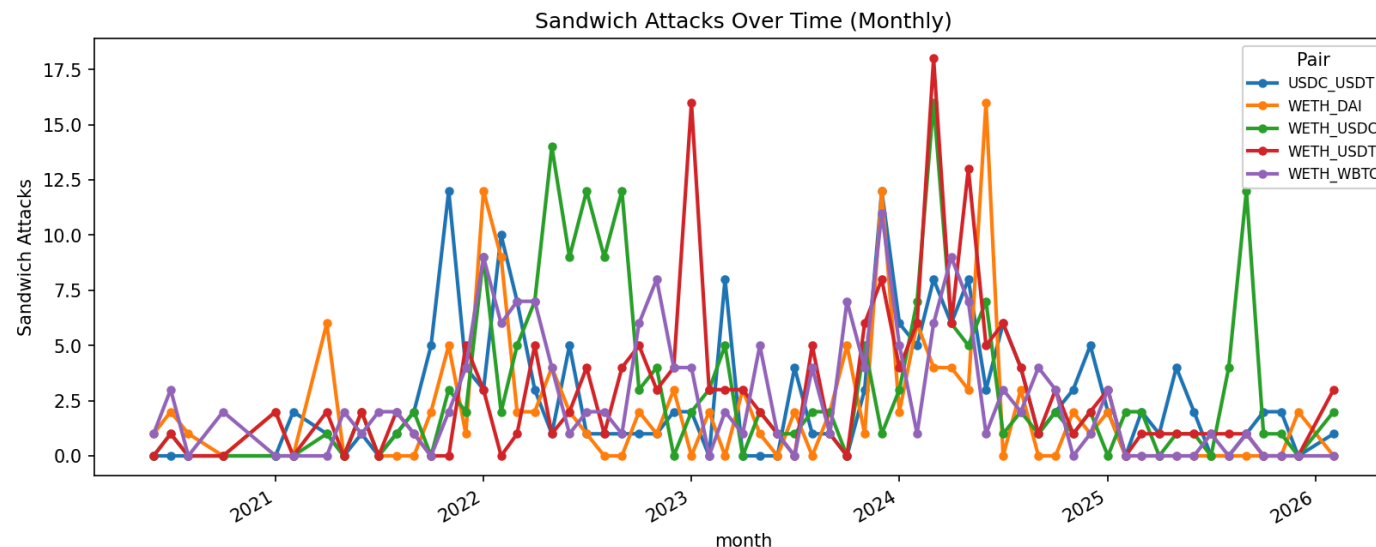


9. Multi-Strategy Entities & Timeline

5 entities operate across **all 4 attack types** simultaneously — evidence of sophisticated MEV bots

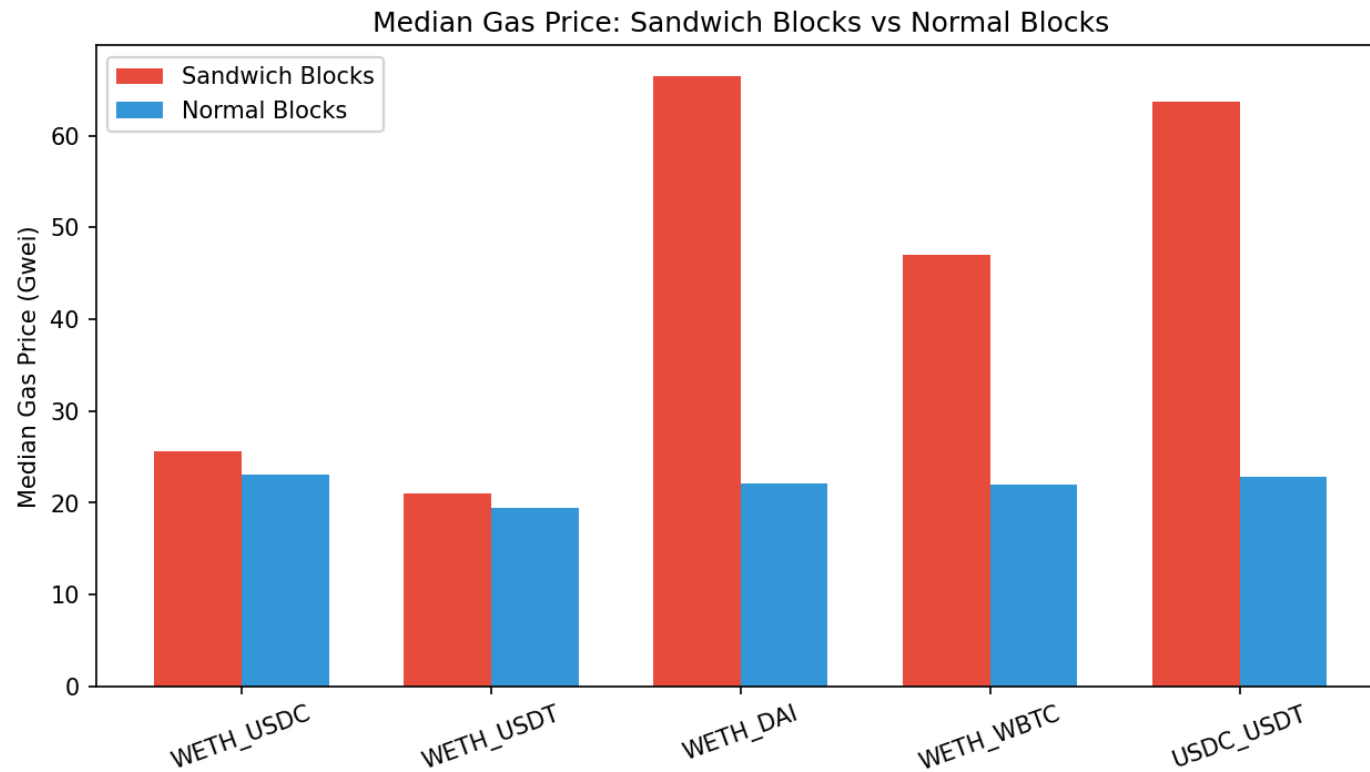
Entity	Strategies
0x00000000...120e49	Sandwich + Displacement + Arbitrage + Suppression
0x6b75d8af...009a80	Sandwich + Displacement + Arbitrage + Suppression
0x3328f7f4...309c49	Sandwich + Displacement + Arbitrage + Suppression

Sandwich timeline: peak activity in **March 2024** (52 events/month), declining afterward as Flashbots/MEV-Boost shifted MEV extraction off-chain



10. Market impact

- Sandwich-related blocks show **significantly higher** median gas prices
- **WETH_DAI, WETH_WBTC, USDC_USDT**: large gas premiums in suspicious blocks
- MEV activity increases gas costs for **all users** in affected blocks
- Entity `0x6b75...80` appears in both back-running and suppression, suggesting **multi-strategy** operation



11. Project Summary

- Complete pipeline: **collection -> detection -> profit analysis -> visualization**
- Covers all **Project 8 requirements**: Displacement, Insertion, Suppression + bonus Arbitrage
- **5,214 events** detected, **\$10.7M+** aggregate adversary profit quantified
- Gas-price analysis confirms MEV raises costs for normal traders

Deliverables

- Python source code (fetcher + analysis pipeline)
- Collected datasets (CSV) + 9 analysis charts + Excel workbook
- Markdown report + GitHub repo for reproducibility

Thank You